

Pearson Edexcel International GCSE

Mathematics A

Modular Unit 2

June 2025 Reconstructed Past Paper

DETAILED SOLUTIONS

Original question followed by a full worked solution

31 working units 106 marks

Selected from Linear Higher Papers 1H and 2H

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L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

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About this reconstructed paper

Every working unit appears in exactly the same order as the Student book. The complete source question appears first; the next page gives a detailed worked solution and final answer.

Unit ownership rule: a question belongs to Unit 2 when its assessed, new or terminal statement is Unit 2, even when the working uses Unit 1 prerequisites. For example, the region-of-inequalities question remains Unit 2 although drawing and reading straight lines is supporting Unit 1 knowledge.

This is an Elite reconstructed teaching paper selected from official Linear Higher material; it is not an official Pearson Modular examination paper. Source assets are retained for private teaching use.

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- 1 The table gives information about the distances 100 adults travel to work.

Distance (d km)	Frequency
$0 < d \leq 5$	26
$5 < d \leq 10$	40
$10 < d \leq 15$	16
$15 < d \leq 20$	10
$20 < d \leq 25$	8

- (a) Write down the modal class.

.....
(1)

- (b) Work out an estimate for the mean distance.

..... km
(4)

Worked Solution 01

Grouped data: modal class and estimated mean

UNIT 2
5 marks**Modal class**

The greatest frequency is 40, so $5 < d \leq 10$.

Estimated mean

$$\bar{d} = \frac{2.5(26) + 7.5(40) + 12.5(16) + 17.5(10) + 22.5(8)}{100} = \frac{920}{100} = 9.2.$$

Final answer

$$5 < d \leq 10, \quad \bar{d} = 9.2 \text{ km}$$

- 2 Anna makes cups.
Each cup costs 6 Swiss francs to make.

Anna puts the cups into boxes to sell.
Each box contains 4 cups.

Anna sells 80 boxes of cups for a total of 2160 Swiss francs.

- (a) Work out the percentage profit Anna makes.
Show your working clearly.

..... %
(4)

Worked Solution 02

Percentage profit

UNIT 2**4 marks****Cost**

$$80 \times 4 \times 6 = 1920 \text{ Swiss francs.}$$

Profit and percentage

$$2160 - 1920 = 240, \quad \frac{240}{1920} \times 100 = 12.5\%.$$

Final answer

12.5%

3

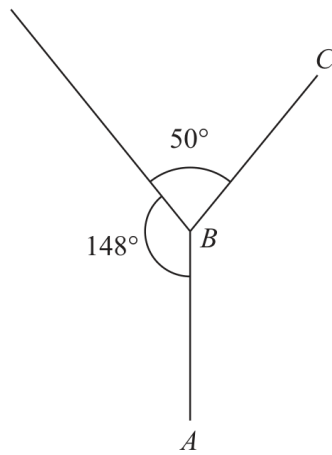


Diagram **NOT**
accurately drawn

AB and BC are two sides of a regular polygon with n sides.

Work out the value of n

Show your working clearly.

$n = \dots\dots\dots$

(Total for Question 3 is 4 marks)

Worked Solution 03

Interior and exterior angles of a regular polygon

UNIT 2**4 marks****Interior angle**

$$\angle ABC = 360^\circ - (148^\circ + 50^\circ) = 162^\circ.$$

Exterior angle and number of sides

$$e = 180^\circ - 162^\circ = 18^\circ, \quad n = \frac{360^\circ}{18^\circ} = 20.$$

Final answer

$$n = 20$$

- 5 In a sale, normal prices are reduced by 28%
The sale price of a watch is 198 euros.
Work out the normal price of the watch.

..... euros

(Total for Question 5 is 3 marks)

Worked Solution 04

Recover a normal price after a percentage reduction

UNIT 2**3 marks****Remaining percentage**

$$100\% - 28\% = 72\%, \quad 0.72N = 198.$$

Solve

$$N = \frac{198}{0.72} = 275.$$

Final answer

275 euros

7 Here is a solid cylinder.

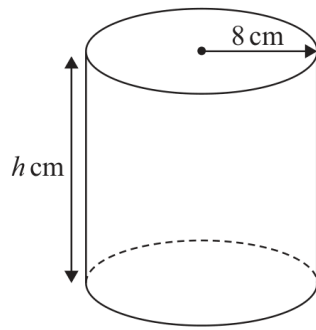


Diagram **NOT**
accurately drawn

The radius of the cylinder is 8 cm

The height of the cylinder is h cm

The volume of the cylinder is 3892 cm^3

Work out the value of h

Give your answer correct to one decimal place.

$h = \dots\dots\dots$

(Total for Question 7 is 3 marks)

Worked Solution 05

Volume of a cylinder

UNIT 2**3 marks****Use the volume formula**

$$3892 = \pi(8)^2 h = 64\pi h.$$

Solve and round

$$h = \frac{3892}{64\pi} = 19.357219 \dots \Rightarrow h = 19.4 \text{ cm (1 d.p.)}.$$

Final answer

$$h = 19.4 \text{ cm}$$

8 (a) Write 520 million in standard form.

.....
(1)

(b) Write 8.79×10^{-5} as an ordinary number.

.....
(1)

Worked Solution 06

Standard form: two conversions

UNIT 2
2 marks**Large number**

$$520\,000\,000 = 5.2 \times 10^8.$$

Small number

$$8.79 \times 10^{-5} = 0.0000879.$$

Final answer

$$5.2 \times 10^8, \quad 0.0000879$$

- (c) Work out $(5 \times 10^{42}) \times (7 \times 10^{-180})$
Give your answer in standard form.

.....
(2)

Worked Solution 07

Multiplication in standard form

UNIT 2**2 marks****Multiply coefficients and powers**

$$(5 \times 10^{42})(7 \times 10^{-180}) = 35 \times 10^{-138} = 3.5 \times 10^{-137}.$$

Final answer

$$3.5 \times 10^{-137}$$

13 Charlotte bought an apartment for \$750 000

In the first year, the value of the apartment decreased by 4%

In the second year, the value of the apartment decreased by 6.5%

In the third year, the value of the apartment increased by $x\%$

At the end of the third year, the value of Charlotte's apartment was \$698 445

Work out the value of x

$x = \dots\dots\dots$

(Total for Question 13 is 3 marks)

Worked Solution 08

Successive depreciation followed by growth

UNIT 2
3 marks**Apply the two decreases**

$$750000(1 - 0.04)(1 - 0.065) = 750000(0.96)(0.935) = 673200.$$

Apply the unknown increase

$$673200 \left(1 + \frac{x}{100}\right) = 698445, \quad 1 + \frac{x}{100} = \frac{698445}{673200} = \frac{83}{80} = 1.0375.$$

Find x

$$x = 3.75.$$

Final answer

$$x = 3.75\%$$

16 $f(x) = 9 - \sqrt{x}$ where $x \geq 0$

$g(x) = 4x^2$

(a) Find $f(9)$

(1)

Worked Solution 09

Function substitution

UNIT 2**1 marks****Substitute**

$$f(9) = 9 - \sqrt{9} = 9 - 3 = 6.$$

Final answer

$$f(9) = 6$$

16 $f(x) = 9 - \sqrt{x}$ where $x \geq 0$
 $g(x) = 4x^2$

- (b) Solve $fg(x) < 0$
Show clear algebraic working.

.....
(3)

Worked Solution 10

Composite function inequality

UNIT 2
3 marks**Compose the functions rigorously**

$$fg(x) = f(4x^2) = 9 - \sqrt{4x^2} = 9 - 2|x|.$$

Solve the inequality as written

$$9 - 2|x| < 0 \Rightarrow |x| > \frac{9}{2} \Rightarrow x < -\frac{9}{2} \text{ or } x > \frac{9}{2}.$$

Read the domain statement carefully

The printed $x \geq 0$ is the input domain of f . Since $g(x) = 4x^2 \geq 0$ for every real x , it does not remove the negative branch.

If the original variable is additionally restricted

If the original variable is also restricted to $x \geq 0$, then $|x| = x$, so the result reduces to $x > \frac{9}{2}$.

Final answer

$$\text{As written: } x < -\frac{9}{2} \text{ or } x > \frac{9}{2}.$$

$$\text{With an additional } x \geq 0 \text{ restriction: } x > \frac{9}{2}.$$

18 y is inversely proportional to x^3

$$y = 4 \text{ when } x = 3$$

Work out the value of x when $y = 864$

$$x = \dots\dots\dots$$

(Total for Question 18 is 4 marks)

Worked Solution 11

Inverse proportionality

UNIT 2**4 marks****Find the constant**

$$y = \frac{k}{x^3}, \quad 4 = \frac{k}{3^3} \Rightarrow k = 108.$$

Use y=864

$$864 = \frac{108}{x^3} \Rightarrow x^3 = \frac{1}{8} \Rightarrow x = \frac{1}{2}.$$

Final answer

$$x = \frac{1}{2}$$

19 The equation of a curve is $y = f(x)$

There is only one minimum point on the curve.

The coordinates of this minimum point are $(8, -12)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x) + 3$

(.....,)
(1)

(ii) $y = f(2x)$

(.....,)
(1)

(Total for Question 19 is 2 marks)

Worked Solution 12

Transformations of a minimum point

UNIT 2**2 marks****Vertical translation**

$$y = f(x) + 3 : \quad (8, -12 + 3) = (8, -9).$$

Horizontal scaling

$$y = f(2x) : \quad 2x = 8 \Rightarrow x = 4, \text{ so } (4, -12).$$

Final answer

$$(8, -9), \quad (4, -12)$$

22 Solve the simultaneous equations

$$\begin{aligned}x^2 + 3y + y^2 &= 7 \\ y &= x + 2\end{aligned}$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

Worked Solution 13

Simultaneous equations with a quadratic substitution

UNIT 2
5 marks**Substitute**

$$y = x + 2 \Rightarrow x^2 + 3(x + 2) + (x + 2)^2 = 7.$$

Solve for x

$$2x^2 + 7x + 3 = 0 = (2x + 1)(x + 3) \Rightarrow x = -\frac{1}{2} \text{ or } -3.$$

Recover y

$$y = x + 2 \Rightarrow y = \frac{3}{2} \text{ or } -1.$$

Final answer

$$(x, y) = \left(-\frac{1}{2}, \frac{3}{2}\right) \text{ or } (-3, -1)$$

23

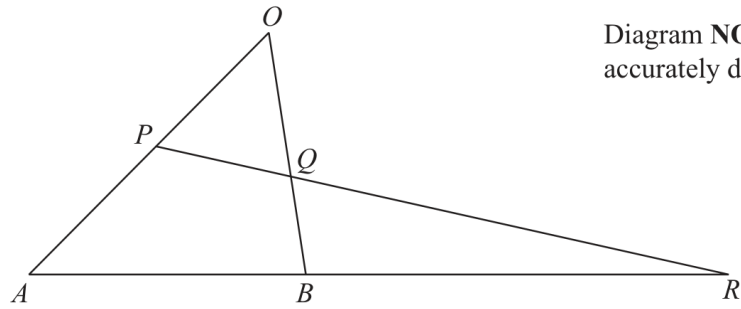


Diagram **NOT**
accurately drawn

OAB is a triangle.

P is the midpoint of OA

Q is a point on OB

ABR and PQR are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

(a) Express $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

.....
(1)

Continued
4MA1-1H Q23

ORIGINAL QUESTION

23

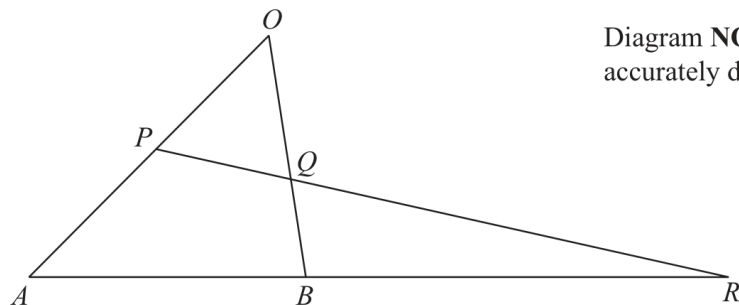


Diagram **NOT**
accurately drawn

OAB is a triangle.

P is the midpoint of OA

Q is a point on OB

ABR and PQR are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

$$AB:BR = 1:2 \quad \vec{OQ} = n\mathbf{b}$$

(b) Use a vector method to find the value of n

$$n = \dots\dots\dots (4)$$

Worked Solution 14

Vector method on two straight lines

UNIT 2
5 marks**Find AB**

$$\vec{AB} = \vec{OB} - \vec{OA} = 8\mathbf{b} - 12\mathbf{a}.$$

Find PR

$$\vec{OR} = \vec{OA} + 3\vec{AB} = -24\mathbf{a} + 24\mathbf{b}, \quad \vec{OP} = 6\mathbf{a}, \quad \vec{PR} = -30\mathbf{a} + 24\mathbf{b}.$$

Use collinearity

$$\vec{OQ} = \vec{OP} + \lambda\vec{PR} = 6\mathbf{a} + \lambda(-30\mathbf{a} + 24\mathbf{b}). \quad 6 - 30\lambda = 0 \Rightarrow \lambda = \frac{1}{5}, \quad n = 24\lambda = \frac{24}{5}.$$

Final answer

$$n = \frac{24}{5}$$

24 An arithmetic series has 30 terms.

The first term is a

The common difference is d

The 20th term is 123

The sum of the 30 terms is 2880

Work out the value of a and the value of d

Show clear algebraic working.

$a =$

$d =$

(Total for Question 24 is 5 marks)

Worked Solution 15

Arithmetic series

UNIT 2**5 marks****20th term**

$$a + 19d = 123.$$

Sum of 30 terms

$$S_{30} = \frac{30}{2}(2a + 29d) = 2880 \Rightarrow 2a + 29d = 192.$$

Solve the pair

$$(2a + 38d) - (2a + 29d) = 246 - 192 \Rightarrow d = 6, \quad a = 9.$$

Final answer

$$a = 9, \quad d = 6$$

- 1 Rio has six cards.
He writes a number on each card so that

the range of the numbers is 10
the median of the numbers is 7.5
the mode of the numbers is 6

Rio arranges the cards so that the numbers are in order of size.

		6		10	14
--	--	---	--	----	----

Three of the numbers are hidden.

Complete the cards above to show the three numbers that are hidden.

(Total for Question 1 is 3 marks)

Worked Solution 16

Ordered cards: range, median and mode

UNIT 2
3 marks**Complete the ordered list**

The list is 4, 6, 6, 9, 10, 14.

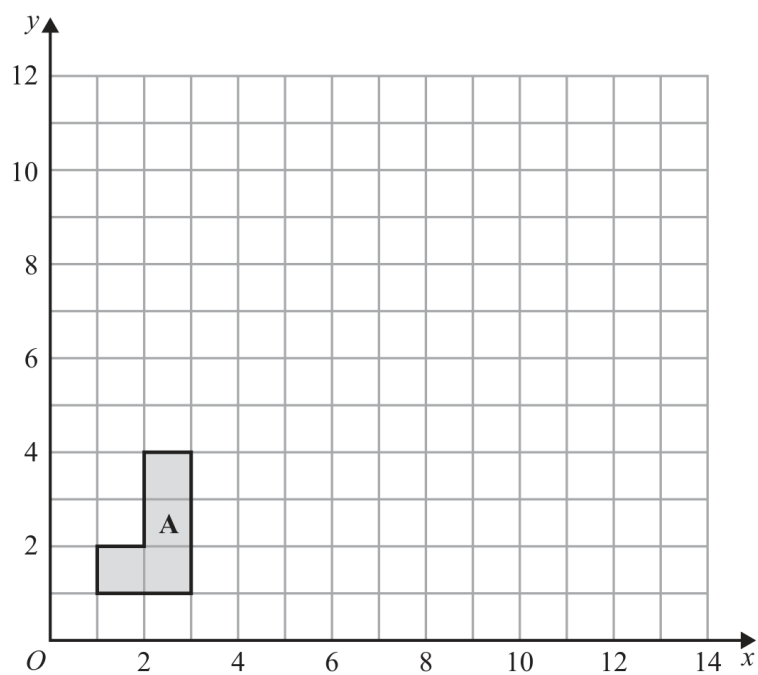
Check the conditions

$$14 - 4 = 10, \quad \frac{6 + 9}{2} = 7.5, \quad \text{mode} = 6.$$

Final answer

4, 6, 9 in the three hidden positions

2



(a) On the grid, enlarge shape A with scale factor 3 and centre $(0, 1)$

(2)

Worked Solution 17

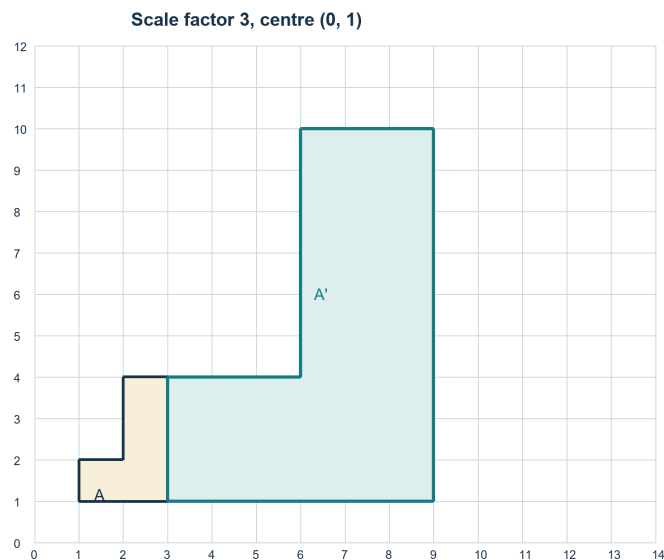
Enlargement from a centre

UNIT 2
2 marks**Apply the map**

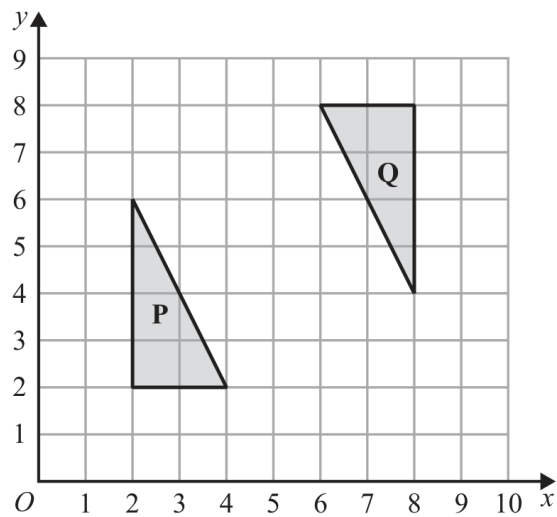
$$(x, y) \mapsto (3x, 3y - 2) \text{ for scale factor 3 about } (0, 1).$$

Image vertices

$$(3, 1), (9, 1), (9, 10), (6, 10), (6, 4), (3, 4).$$

**Final answer**

The enlarged shape has vertices $(3, 1), (9, 1), (9, 10), (6, 10), (6, 4), (3, 4)$



(b) Describe fully the single transformation that maps triangle **P** onto triangle **Q**

.....

.....

(3)

Worked Solution 18

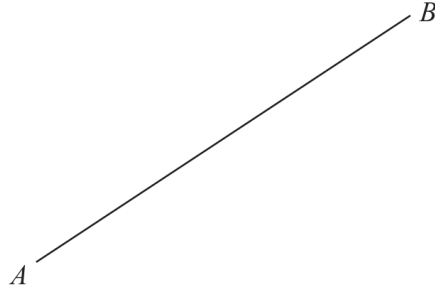
Rotation of a triangle

UNIT 2**3 marks****Identify the transformation** $P \rightarrow Q$ is a 180° rotation about $(5, 5)$.**Check one vertex**

$$(2, 2) \mapsto (2 \cdot 5 - 2, 2 \cdot 5 - 2) = (8, 8), \text{ matching } Q.$$

Final answerRotation 180° about $(5, 5)$

- 4 Using ruler and compasses only, construct the perpendicular bisector of the line AB .
Show all your construction lines.



(Total for Question 4 is 2 marks)

Worked Solution 19

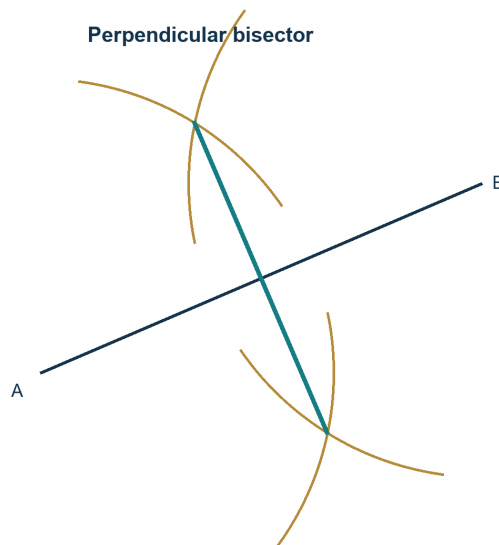
Perpendicular bisector construction

UNIT 2
2 marks**Construct arcs**

With compass radius greater than half AB , draw intersecting arcs from A and B on both sides.

Draw the bisector

Join the two arc intersections; this line is perpendicular to AB and bisects it.

**Final answer**

A correct perpendicular bisector with two pairs of construction arcs

- 5 The diagram shows two similar quadrilaterals, $ABCD$ and $EFGH$

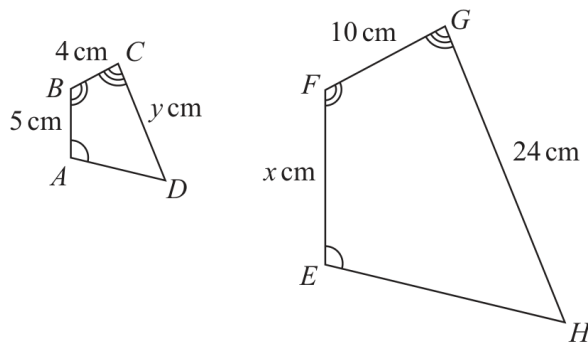


Diagram **NOT**
accurately drawn

$$AB = 5 \text{ cm} \quad BC = 4 \text{ cm} \quad CD = y \text{ cm}$$

$$EF = x \text{ cm} \quad FG = 10 \text{ cm} \quad GH = 24 \text{ cm}$$

- (a) Work out the value of x

$$x = \dots\dots\dots (2)$$

Continued
4MA1-2H Q5

ORIGINAL QUESTION

- 5 The diagram shows two similar quadrilaterals, $ABCD$ and $EFGH$

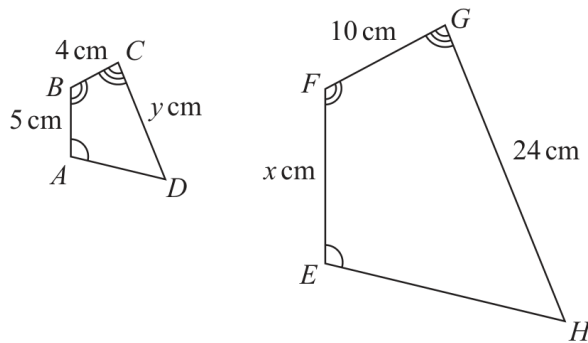


Diagram **NOT**
accurately drawn

$$\begin{array}{lll} AB = 5 \text{ cm} & BC = 4 \text{ cm} & CD = y \text{ cm} \\ EF = x \text{ cm} & FG = 10 \text{ cm} & GH = 24 \text{ cm} \end{array}$$

- (b) Work out the value of y

$$y = \dots\dots\dots (2)$$

Worked Solution 20

Similar quadrilaterals

UNIT 2**4 marks****Scale factor**

$$\frac{FG}{BC} = \frac{10}{4} = 2.5.$$

Find x and y

$$x = 5(2.5) = 12.5, \quad y = \frac{24}{2.5} = 9.6.$$

Final answer

$$x = 12.5 \text{ cm}, \quad y = 9.6 \text{ cm}$$

- 6 Pau, Sam and Tia share £240 in the ratios 3 : 4 : 5

Sam and Tia each give £10 of their share to Pau.

Work out the ratios of the amounts of money that Pau, Sam and Tia now have.
Give your answer in its simplest form.

..... : :

(Total for Question 6 is 4 marks)

Worked Solution 21

Ratio after transfers

UNIT 2
4 marks**Initial shares**

$$240 \div (3 + 4 + 5) = 20 \text{ per part; } (P, S, T) = (60, 80, 100).$$

Transfers

$$(P, S, T) = (60 + 10 + 10, 80 - 10, 100 - 10) = (80, 70, 90).$$

Simplify

$$80 : 70 : 90 = 8 : 7 : 9.$$

Final answer

$$8 : 7 : 9$$

- 7 Tim buys a bracelet for 4000 Swiss francs.
The value of the bracelet increases by 7% each year.

Work out the value of the bracelet at the end of 3 years.
Give your answer correct to the nearest Swiss franc.

..... Swiss francs

(Total for Question 7 is 3 marks)

Worked Solution 22

Compound percentage growth

UNIT 2**3 marks****Three annual increases**

$$4000(1.07)^3 = 4900.172 \dots$$

Round

To the nearest Swiss franc: 4900.

Final answer

4900 Swiss francs

8 Solve the simultaneous equations

$$\begin{aligned}3x + 5y &= 8 \\4x + y &= -3.5\end{aligned}$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 8 is 3 marks)

Worked Solution 23

Simultaneous linear equations

UNIT 2

3 marks

Eliminate x

$$5(4x + y) = -17.5 \Rightarrow 20x + 5y = -17.5; \quad (20x + 5y) - (3x + 5y) = -17.5 - 8.$$

Solve

$$17x = -25.5 \Rightarrow x = -1.5, \quad y = -3.5 - 4(-1.5) = 2.5.$$

Final answer

$$x = -1.5, \quad y = 2.5$$

9 (a) Solve the inequality $7 - 3t < 2t + 15$

(2)

Worked Solution 24

Linear inequality

UNIT 2**2 marks****Collect terms**

$$7 - 3t < 2t + 15 \Rightarrow -5t < 8.$$

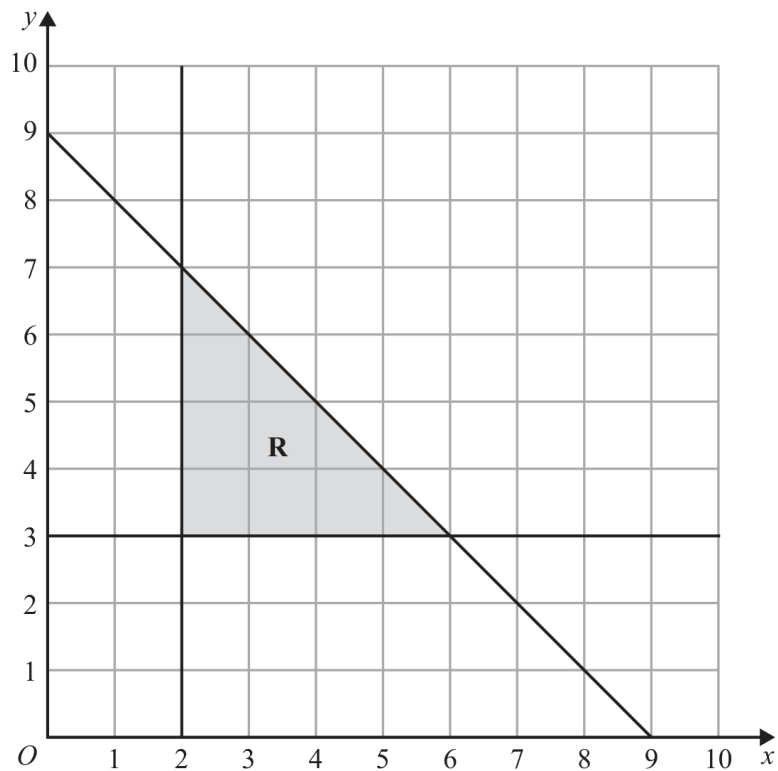
Reverse on division

$$t > -\frac{8}{5} = -1.6 \quad \text{because division is by } -5.$$

Final answer

$$t > -1.6$$

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



(b) Write down three inequalities that define the region **R**

.....
.....
.....
(3)

Worked Solution 25

Region of inequalities

UNIT 2**3 marks****Read the three boundaries**

$$x = 2, \quad y = 3, \quad x + y = 9.$$

Choose the shaded side

$$R : \quad x \geq 2, \quad y \geq 3, \quad x + y \leq 9.$$

Final answer

$$x \geq 2, \quad y \geq 3, \quad x + y \leq 9$$

- 13 The frequency table gives information about the times, in minutes, that 60 people took to complete a puzzle.

Time (t minutes)	Frequency
$0 < t \leq 10$	8
$10 < t \leq 20$	13
$20 < t \leq 30$	12
$30 < t \leq 40$	17
$40 < t \leq 50$	7
$50 < t \leq 60$	3

- (a) Complete the cumulative frequency table.

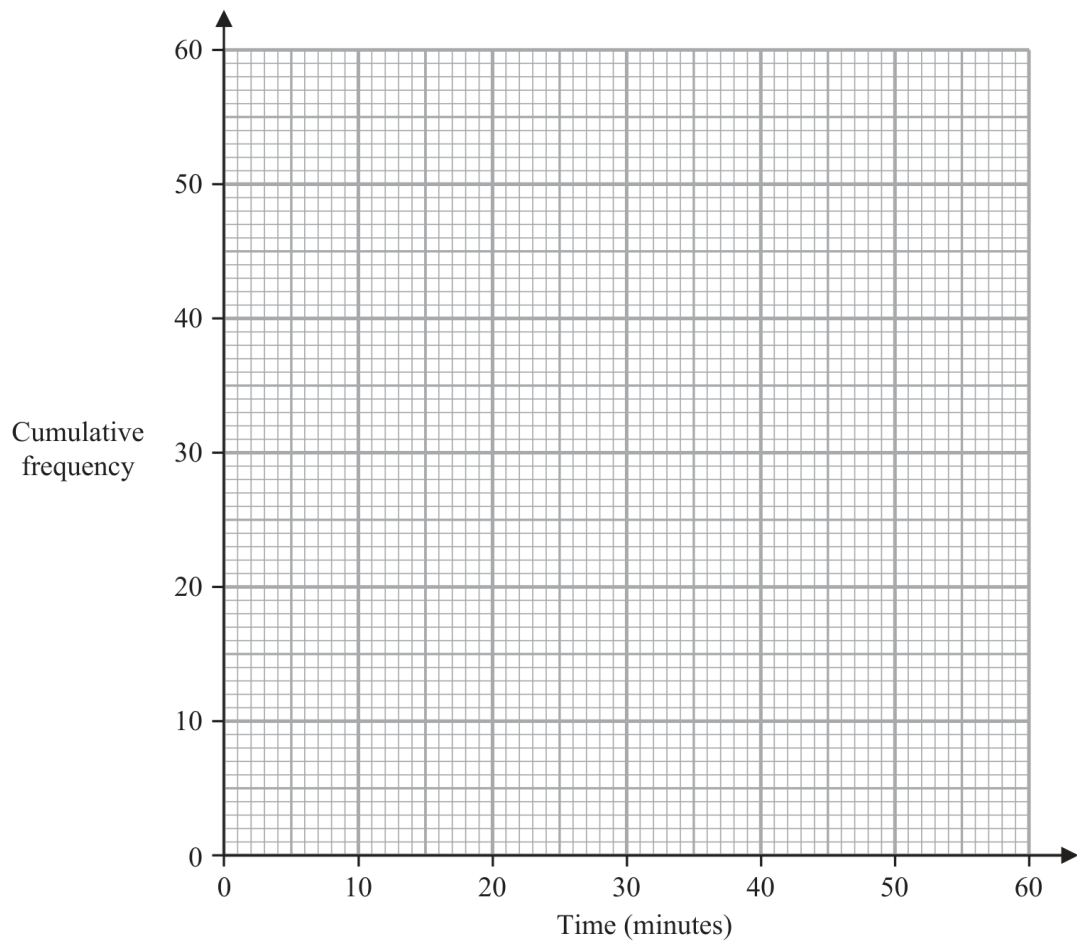
Time (t minutes)	Cumulative frequency
$0 < t \leq 10$	
$0 < t \leq 20$	
$0 < t \leq 30$	
$0 < t \leq 40$	
$0 < t \leq 50$	
$0 < t \leq 60$	

(1)

- (b) On the grid opposite, draw a cumulative frequency graph for your table.

Continued
4MA1-2H Q13

ORIGINAL QUESTION



(2)

- (c) Use your graph to find an estimate for the median time.

..... minutes
(1)

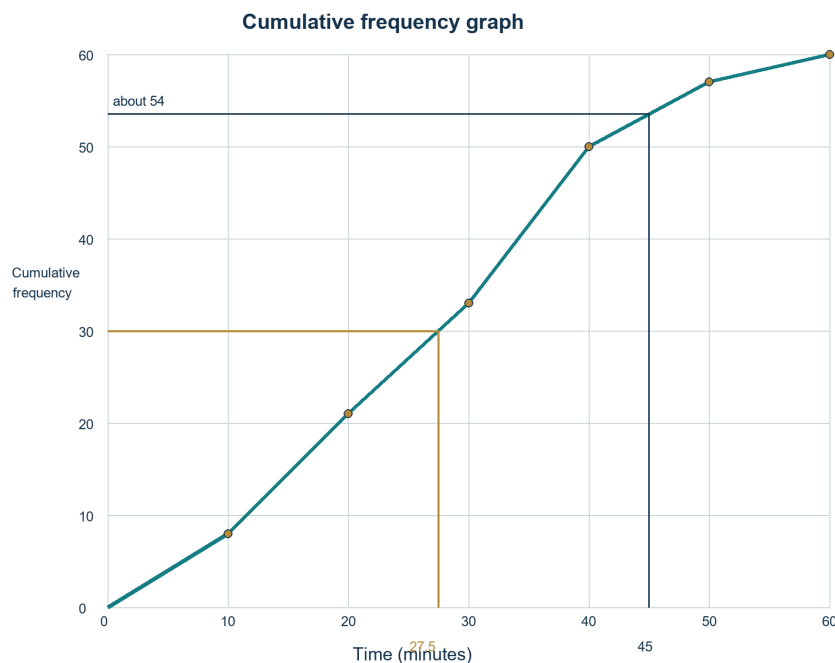
- (d) Use your graph to find an estimate for the number of these people who took more than 45 minutes to complete the puzzle.

.....
(2)

(Total for Question 13 is 6 marks)

Worked Solution 26

Cumulative frequency graph and estimates

UNIT 2
6 marks**Cumulative totals** $8, 8 + 13 = 21, 33, 50, 57, 60.$ **Plot** $(10, 8), (20, 21), (30, 33), (40, 50), (50, 57), (60, 60)$, joined as a cumulative-frequency graph.**Read estimates**Median: 27 to 28 minutes; more than 45 minutes: $60 - 54 \approx 6$ people.**Final answer**

8, 21, 33, 50, 57, 60; median 27 to 28 min; about 6 people

16 Make t the subject of the formula $c = \frac{t^2 + 3}{7 - 8t^2}$

(Total for Question 16 is 4 marks)

Worked Solution 27Make t the subject**UNIT 2****4 marks****Clear the denominator**

$$c(7 - 8t^2) = t^2 + 3 \Rightarrow 7c - 8ct^2 = t^2 + 3.$$

Collect and factor

$$7c - 3 = (1 + 8c)t^2.$$

Take square roots

$$t^2 = \frac{7c - 3}{1 + 8c} \Rightarrow t = \pm \sqrt{\frac{7c - 3}{1 + 8c}}.$$

Final answer

$$t = \pm \sqrt{\frac{7c - 3}{1 + 8c}}$$

17 $y = 4x^3 + 5x^2 + 2x$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots (2)$$

(b) Find the coordinates of the turning points on the graph with equation $y = 4x^3 + 5x^2 + 2x$
Show clear algebraic working.

$$\dots\dots\dots (4)$$

(Total for Question 17 is 6 marks)

Worked Solution 28

Derivative and turning points

UNIT 2
6 marks**Differentiate**

$$\frac{dy}{dx} = 12x^2 + 10x + 2.$$

Solve derivative equal to zero

$$12x^2 + 10x + 2 = 2(3x + 1)(2x + 1) = 0 \Rightarrow x = -\frac{1}{3}, -\frac{1}{2}.$$

Substitute

$$x = -\frac{1}{2} \Rightarrow y = -\frac{1}{4}; \quad x = -\frac{1}{3} \Rightarrow y = -\frac{7}{27}.$$

Final answer

$$\frac{dy}{dx} = 12x^2 + 10x + 2; \quad \left(-\frac{1}{2}, -\frac{1}{4}\right), \left(-\frac{1}{3}, -\frac{7}{27}\right)$$

- 19** Solve the inequality $4x^2 + 4x - 15 < 0$
Show clear algebraic working.

.....
(Total for Question 19 is 3 marks)

Worked Solution 29

Quadratic inequality

UNIT 2**3 marks****Factorise**

$$4x^2 + 4x - 15 = (2x - 3)(2x + 5).$$

Use the sign of an upward quadratic

The expression is negative between the roots $-\frac{5}{2}$ and $\frac{3}{2}$.

Final answer

$$-2.5 < x < 1.5$$

21 The diagram shows a cuboid.

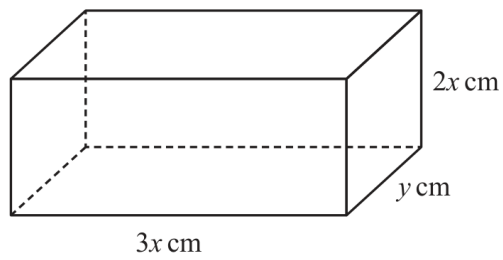


Diagram **NOT**
accurately drawn

The cuboid measures $3x$ cm by $2x$ cm by y cm

The volume of the cuboid is 1014 cm^3

The total surface area of the cuboid is $A\text{ cm}^2$

Show that $A = 12x^2 + \frac{1690}{x}$

You must show all the stages of your working.

(Total for Question 21 is 3 marks)

Worked Solution 30

Surface area of a cuboid

UNIT 2
3 marks**Use volume**

$$6x^2y = 1014 \Rightarrow y = \frac{169}{x^2}.$$

Write surface area

$$A = 2(6x^2 + 3xy + 2xy) = 12x^2 + 10xy.$$

Substitute

$$A = 12x^2 + 10x \left(\frac{169}{x^2} \right) = 12x^2 + \frac{1690}{x}.$$

Final answer

$$A = 12x^2 + \frac{1690}{x}$$

26 **A** and **B** are two similar solids.

The height of solid **A** is 31 cm

The height of solid **B** is 18.6 cm

Given that

$$\text{volume of solid A} - \text{volume of solid B} = 735 \text{ cm}^3$$

find the volume of solid **A**

..... cm³

(Total for Question 26 is 4 marks)

Worked Solution 31

Volumes of similar solids

UNIT 2
4 marks**Scale volumes**

$$A : B = 31 : 18.6 = 5 : 3 \Rightarrow V_A : V_B = 5^3 : 3^3 = 125 : 27.$$

Use the difference

$$V_A = 125k, V_B = 27k; \quad 98k = 735 \Rightarrow k = 7.5.$$

Find A

$$V_A = 125(7.5) = 937.5 \text{ cm}^3.$$

Final answer

$$937.5 \text{ cm}^3 \text{ (or } 938 \text{ cm}^3\text{)}$$